

Ruitang Yang

Glaciologist | Glacier Modelling, Data Assimilation and Satellite Remote Sensing

University of Oslo · Oslo, Norway | ruitang.yang@geo.uio.no | [ORCID](#) | [Google Scholar](#) | [GitHub](#)

ACADEMIC PROFILE

Glaciologist and researcher specializing in glacier modelling, data assimilation, and satellite remote sensing. Integrates process-based models, Earth-observation data, and Bayesian inference to quantify glacier change and produce uncertainty-aware projections from regional to global scales.

CURRENT APPOINTMENT

2022–pres. **Researcher** | University of Oslo | Norway
Leads development of observation-constrained glacier-modelling and data-assimilation workflows integrating process-based models, Earth-observation data, Bayesian calibration, and ensemble uncertainty analysis.

EDUCATION

2016–2021 **PhD in Physical Geography** | University of Chinese Academy of Sciences | China
Thesis: Glacier Change on the Kenai Peninsula, Alaska, Using Remote Sensing and Modelling.
Visiting researcher at University of Alaska Fairbanks, USA (18 mos.) and University of Oslo, Norway (1 yr.)

2012–2015 **MSc in Mathematics Education** | Shaanxi Normal University | China

2007–2011 **BSc in Mathematics and Applied Mathematics** | Shaanxi Normal University | China

SELECTED RESEARCH CONTRIBUTIONS

- **Marine-terminating glacier modelling.**
Led development of a coupled OGGM–PyGEM–SERMeQ framework integrating climatic mass balance, glacier geometry, ice dynamics, and frontal ablation. Demonstrated for 71 marine-terminating glaciers in Svalbard using joint Bayesian calibration, multiple observational constraints, and uncertainty propagation through 2100.
- **Kenai Peninsula glacier change.**
Developed multi-decadal satellite Earth-observation and modelling workflows to quantify changes in glacier area, mass, surface velocity, and glacier–runoff relationships across the Kenai Peninsula, Alaska, using observations from 1986–2019. The work resulted in three first-author publications and three public regional datasets.

SELECTED PUBLICATIONS

- **Yang, R.,** Ultee, L., Aalstad, K., Debolskiy, M. V., Hock, R., Schmitt, P., Rounce, D., and Li, T. (2026). Joint Bayesian Calibration of Frontal Ablation and Surface Mass Balance in Global Glacier Models. *EGUsphere [preprint]*. [doi:10.5194/egusphere-2026-1081](https://doi.org/10.5194/egusphere-2026-1081)
- **Yang, R.,** Hock, R., Rounce, D. R., and Kang, S. (2025). Regional-scale response of glacier speed to seasonal runoff variations on the Kenai Peninsula, Alaska. *Geophysical Research Letters*, 52. [doi:10.1029/2025GL115248](https://doi.org/10.1029/2025GL115248)
- **Yang, R.,** Hock, R., Kang, S., Guo, W., Shangguan, D., Jiang, Z., and Zhang, Q. (2022). Glacier surface speed variations on the Kenai Peninsula, Alaska, 2014–2019. *Journal of Geophysical Research: Earth Surface*. [doi:10.1029/2022JF006599](https://doi.org/10.1029/2022JF006599)
- **Yang, R.,** Hock, R., Kang, S., Shangguan, D., and Guo, W. (2020). Glacier mass and area changes on the Kenai Peninsula, Alaska, 1986–2016. *Journal of Glaciology*. [doi:10.1017/jog.2020.32](https://doi.org/10.1017/jog.2020.32)
- Aalstad, K., Alonso-González, E., Pirk, N., Westermann, S., Willmes, C., and **Yang, R.** (2026). Evolving beyond collapse: An adaptive particle batch smoother for cryospheric data assimilation. *arXiv preprint arXiv:2601.20049*.

Complete publication list: [Google Scholar](#)

SELECTED RESEARCH SOFTWARE AND DATASETS

- **Coupled Frontal-Ablation Modeling Framework.** Lead designer and developer of a research software framework for observation-constrained modelling of marine-terminating glaciers. [doi:10.5281/zenodo.18761729](https://doi.org/10.5281/zenodo.18761729)
- **Kenai Peninsula research datasets.** Glacier mass and area change (1986–2016), velocity maps (2014–2019), and regional surface-speed products. [PANGAEA](#) and [Mendeley Data](#).

TECHNICAL EXPERTISE

- **Programming and computing:** Python, Shell, Git, Linux, HPC; MATLAB and C
- **Scientific modelling and inference:** OGGM, PyGEM, SERMeQ; process-based modelling, Bayesian inference, data assimilation, ensemble simulation, model calibration, and uncertainty quantification
- **Geospatial and Earth observation:** QGIS, ArcGIS, ENVI, GAMMA; optical, SAR, and laser-altimetry data
- **Research practice:** reproducible workflows, multi-source data integration, model evaluation, open-source software, and research-data publication

AWARDS AND RECOGNITION

- 2024 **Top Cited Article 2022–2023**
Journal of Geophysical Research: Earth Surface
- 2021 **Best Student Oral Presentation**
International Glaciological Society Nordic Branch Symposium | Oslo

SELECTED PROFESSIONAL SERVICE

- 2022–Pres. **Member and subgroup co-chair, GlacierMIP**
Co-leading development of standardized experimental protocols for GlacierMIP4.
- 2023–Pres. **Peer reviewer**
Journal of Glaciology; Advances in Climate Change Research; GIScience & Remote Sensing; AGU OSPA
- 2025 **Organizing committee member**
CryoDA Workshop and Glacier Modelling Workshop | University of Oslo
- 2022–2023 **Member, Working Group on Regional Assessments of Glacier Mass Change (RAGMAC) |**
International Association of Cryospheric Sciences

SELECTED PRESENTATIONS

- 2026 **Modeling Frontal Ablation in Global Glacier Models**
EGU General Assembly | Vienna · PICO
- 2024 **Regional-Scale Response of Glacier Speed to Seasonal Runoff Variations on the Kenai Peninsula, Alaska**
AGU Fall Meeting | Washington, D.C. · Poster
- 2021 **Glacier Changes in the Kenai Peninsula, Alaska: Mass Balance and Ice Flow Using Remote Sensing**
International Glaciological Society Nordic Branch Symposium | Oslo · Oral

TEACHING AND MENTORING

- 2011–2016 **Senior High School Mathematics Teacher and Class Adviser**
Longxi No. 1 Middle School | China
Five years of mathematics teaching; served for three years as class adviser with responsibility for approximately 60 students.
- 2016–2018 **Co-mentor, MSc thesis in glacier remote sensing**
University of Chinese Academy of Sciences | China

SELECTED FIELD EXPERIENCE

- 2018 & 2020 **Field assistant, Kennicott Glacier**
Alaska, USA
Ablation-stake drilling, automatic weather station and time-lapse camera installation, and snow/water sampling; four weeks across two campaigns.
- 2019 **Field assistant, Halong Glacier**
Qinghai–Tibet Plateau, China
Ablation-stake drilling, automatic weather station installation, and glacier-flow measurements; three weeks.
- 2017 **Snow and UAV field campaign**
Zhangjiakou, China
Snow-pit measurements, melt-stake drilling, time-lapse camera installation, and UAV surveys; 66 days.

SELECTED ADVANCED TRAINING

- 2023 **Summer School on Data Assimilation and Applications**
TU Delft and NORCE | Romania
- 2022 **Machine Learning in Glaciology Workshop**
University of Oslo, UC Berkeley, and Utrecht University | Finse, Norway
- 2022 **OGGM–PyGEM Global Glacier Modelling Workshop**
University of Oslo, University of Bremen, and University of Innsbruck | Finse, Norway